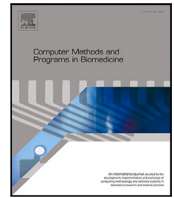




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SD-LayerNet: Robust and label-efficient retinal layer segmentation via anatomical priors

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ABSTRACT

Background and objectives: Automated, anatomically coherent retinal layer segmentation in optical coherence tomography (OCT) is one of the most important components of retinal disease management. However, current methods rely on large amounts of labeled data, which can be difficult and expensive to obtain. In addition, these systems tend often propose anatomically impossible results, which undermines their clinical reliability.**Methods:** This study introduces a semi-supervised approach to retinal layer segmentation that leverages large amounts of unlabeled data and anatomical prior knowledge related to the structure of the retina. During training, we use a novel topological engine that converts inferred retinal layer boundaries into pixel-wise structured segmentations. These compose a set of anatomically valid disentangled representations which, together with predicted style factors, are used to reconstruct the input image. At training time, the retinal layer boundaries and pixel-wise predictions are both guided by reference annotations, where available, but more importantly by innovatively exploiting anatomical priors that improve the performance, robustness and coherence of the method even if only a small amount of labeled data is available.**Results:** Exhaustive experiments with respect to label efficiency, contribution of unsupervised data and robustness to different acquisition settings were conducted. The proposed method showed state-of-the-art performance on all the studied public and internal datasets, specially in low annotated data regimes. Additionally, the model was able to make use of unlabeled data from a different domain with only a small performance drop in comparison to a fully-supervised setting.**Conclusion:** A novel, robust, label-efficient retinal layer segmentation method was proposed. The approach has shown state-of-the-art layer segmentation performance with a fraction of the training data available, while at the same time, its robustness against domain shift was also shown.

1. Introduction

Optical Coherence Tomography (OCT) is the primary imaging modality used to monitor retinal diseases and manage patients. This imaging technique allows ophthalmologists a cross-sectional view of the retina and to evaluate a variety of imaging biomarkers, such as the thickness of the retinal layers. These biomarkers are important indicators of conditions such as Age-related Macular Degeneration (AMD), and diabetic macular edema (DME), two major causes of blindness worldwide [1–5]. However, the large volume of data that is collected in clinical routine and the heavy workload of clinicians make

manual layer delineation (Fig. 1) infeasible in a busy clinical setting. As a result, automated retinal layer segmentation methods started to be developed and offered by the scanner manufacturers as soon as OCT became commercially and widely available [6].

Current state-of-the-art retinal layer segmentation algorithms often rely on supervised deep learning, i.e. they require training with large amounts of annotated images. However, obtaining high-quality manual annotations of pathological cases is a complex, tiresome, and costly process. Instead, novel solutions should be *label-efficient*, reducing the amount of annotations required by exploiting the complementary information available in the often abundant unlabeled data resulting

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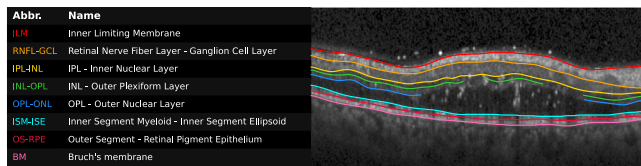


Fig. 1. Overview of the retinal layer boundaries and nomenclature used in this study, illustrated on a sample image from the DukeDME dataset [7]. At some locations, the position of the boundaries could not be determined.

from clinical routine. In particular, OCT retinal segmentation benefits from the fact that retinal layers and pathomorphological manifestations occur in a structured and topologically consistent fashion, with layers having always the same relative position.

This *anatomical prior knowledge* can facilitate the deep learning process if exploited properly.

Another pitfall of common deep learning approaches is that they tend to fail for cases outside their training domain, most of the time providing overconfident and yet erroneous predictions. When translating to clinical practice, this can not only have direct consequences on patient diagnosis and follow-up, but can also undermine the confidence of clinicians in such systems. Therefore, retinal layer segmentation systems should be *reliable* and *robust*. For that, there should be a focus on reducing the amount of anatomically implausible and incoherent segmentations, even when operating in new domains.

While semi-supervised methods have demonstrated promise in reducing label requirements, they often overlook the incorporation of anatomical priors and provide limited guarantees for anatomical coherence. This can lead to biologically implausible predictions, particularly in complex or pathological cases.

In this paper, we address both the *label efficiency* and the *robustness* aspects by proposing a semi-supervised retinal layer segmentation model that leverages *anatomical priors* to enhance performance under label scarcity. It builds upon the concept of *disentangled representations*, which enables it to separate anatomically relevant spatial factors from image domain-specific style factors of retinal OCT scans. We extensively evaluated the properties and the generalization of our layer segmentation method on a number of public and private datasets covering OCT cases acquired with different scanning devices and representative of three different retinal diseases.

1.1. Related works

Retinal layer segmentation. Early approaches for retinal layer segmentation involved extracting features from B-scans (cross-sectional images of the retina) and using graph-based methods to estimate the surface positions. For example, the IOWA Reference Algorithms [8,9] represented the OCT data as a graph and solved for the surface positions using graph-search algorithms, while also enforcing correct topological ordering, prior layer thickness constraints, and smoothness. These graph-based methods were further improved in [10–12]. Rathke et al. [13] proposed a method that used a probabilistic graphical model and incorporated anatomical shape priors for OCT segmentation, including post-processing to fix errors in the boundary maps. However, these methods rely on hand-crafted image features and may perform poorly in the presence of noise or other imaging artifacts, or in cases of severe pathologies. Some approaches have attempted to improve on these limitations by using machine learning to estimate the cost function for the nodes of the graph [14–17], but they still rely on the quality of the initial probability map and are subject to predefined constraints on layer thickness and smoothness.

With the emergence of deep learning, U-Net [18] and its variants became a popular approach for retinal layer segmentation. For instance, ReLayNet [19] predicts the pixel-wise locations of nine retinal layers

and possible fluid-filled pockets. The MGU-Net model [20] improves upon this by introducing a two-stage segmentation and a graph reasoning module, reducing both the model size and training time, while offering better performance. The UNeXt model [21] uses convolutional multilayer perceptrons for rapid medical image segmentation, while also predicting pixel-wise segmentation. However, these algorithms do not guarantee a single unique boundary position in each A-scan (a column in a B-scan). To address this, TCCT-BP [22] introduces a hybrid transformer-CNN based method for layer retinal segmentation with an extra convolutional step trained to match the vertical gradients of the pixel based segmentations with that of the manual reference. However, this approach still does not provide a strict guarantee of proper anatomical order.

Despite their merits, none of these methods considers prior anatomical knowledge, e.g. the natural ordering of the layers, which can lead to anatomically implausible results. He et al. [23] addressed this issue by predicting surface positions using column-wise soft-argmax, ensuring that only one position is inferred per A-scan, while also including a topological module to guarantee proper layer ordering. The model considers the 1D boundary regression task and the 2D layer segmentation task independently by having two output branches. Because of this, these tasks do not directly influence each other. The BAU-Net [24] employs a similar topological module with layer order guarantees. It contains a special U-Net variant with edge-aware modules in the skip connections, with a special Canny Edge Fusion module, which combines the pixel-wise layer segmentations with the boundary regressions in a more connected manner. However, only the segmentation task can influence the regression task, not vice versa, since the Canny Edge fusion is not differentiable. Besides not allowing the flow of gradients from the regression task to the segmentation task, it also does not allow posing topological guarantees on the segmentation task. The transformer for retinal layer segmentation (TRLS) [25] uses a transformer-based architecture to directly regress the retinal layer boundaries, without an intermediate 2D representation, while maintaining the correct topological order of the layers.

Several methods have been developed to utilize the 3D information available in volumes [26–30], demonstrating that this additional spatial information enhances segmentation compared to methods that work solely on 2D B-scans. Nevertheless, due to the misalignment of adjacent B-scans caused by involuntary head or eye movements, most fully 3D methods require the flattening of volumes to remove motion artifacts. This flattening process, however, necessitates a preliminary layer segmentation step during pre-processing. Also, these methods are typically more computationally demanding. Interestingly, recent studies [29,30] have shown that the 3D aggregation of the B-scan-wise predictions from a 2D segmentation model can outperform a full 3D model. Because of both aforementioned limitations of 3D approaches and these recent findings, we do not consider 3D segmentation methods in this manuscript and instead focus on producing high quality B-scan-level predictions.

Label efficiency. The current state-of-the-art deep-learning-based retinal layer segmentation requires a large amount of labeled training data that can be difficult and costly to obtain. To address this challenge, some approaches use semi-supervised learning, which leverages the abundance of unlabeled data to complement the scarce labeled data. One common technique for deep semi-supervised learning involves the integration of a segmentation network, generating initial segmentations. Subsequently, a second network is employed to assess the realism and alignment of these predictions with the labeled data, and through an adversarial loss encourages the segmentation network to produce more realistic outputs [31–33]. However, adversarial learning-based approaches do not explicitly incorporate domain-specific anatomical priors, making them prone to generating segmentations that lack anatomical plausibility, particularly in pathological cases.

Yang et al. [34] proposed a semi-supervised model using both OCT and OCTA data, based on cross-consistency training. In particular, two encoders are trained, one with labeled data, another with unlabeled, and the consistency between the two is maintained during the training. However, this method relies solely on consistency enforcement and does not incorporate structural retinal knowledge or enforce topological guarantees, potentially leading to anatomically inconsistent predictions. BE-SemiNet [35] improves upon this approach by introducing geometric constraints to ensure the relative order of predicted boundary positions. Nevertheless, these constraints are limited to 1D boundary predictions and do not extend to enforcing anatomical consistency across the full 2D segmentation map. Moreover, the constraints are non-differentiable, which restricts their influence on the learning process during training. Additionally, BE-SemiNet treats boundary regression and pixel-wise segmentation as separate tasks, preventing them from mutually improving each other. These limitations reduce its robustness to pathological cases and its ability to produce anatomically coherent results under challenging conditions.

A different technique was employed in the Spatial Decomposition Network (SD-Net) proposed by Chartsias et al. [36], for cardiac image analysis. This method disentangles an image into binary spatial anatomical factors and non-spatial modality-dependent style, which are used to reconstruct the original image. With a small number of labeled instances and an optional adversarial loss function, some of the anatomical factors can be encouraged to represent meaningful anatomical regions. These anatomical factors are subsequently used as input for the segmentation head for the target segmentation task. Through the reconstruction task, this method allows for the use of unlabeled data. While effective for cardiac images, this approach does not integrate retinal-specific anatomical constraints, limiting its applicability to retinal OCT segmentation.

In a preliminary version of this work, we proposed the Spatial Decomposition Layer Segmentation Network (SD-LayerNet) [37], a fully convolutional approach that combines supervised and semi-supervised training for retinal layer segmentation. SD-LayerNet leverages disentangled representations and anatomical priors to achieve state-of-the-art performance while requiring significantly less annotated data for training. In this paper, we substantially expand our prior work by: 1. Introducing a more compact, yet robust set of *anatomical priors* encoded as *loss terms*. In particular, we found that the *surface continuity constraint* and the *probability distribution standard deviation constraint* of the preliminary work did not significantly contribute to the performance, while using a new formulation of the *surface slope constraint* resulted in better performance in highly tilted scans. 2. Proposing a more *anatomically coherent* way of ensuring the correct *topological ordering* of retinal layers, by assuming that in atrophies layer collapse towards the bottom of the retina instead of towards the ILM as in Fazekas et al. [37]. 3. Putting forth a novel method for converting 1D regression results into 2D segmentation maps, directly connecting the regression task with the segmentation task, allowing to enforce strict topological guarantees on the 2D segmentation maps, a feature not present in any of the prior works. 4. Presenting a joint training setup for the tightly coupled 1D boundary regression and the 2D segmentation task, where one task directly influences the other and evaluate its effect by separately investigating the performance by training the network using solely the 1D boundary positions, and by the 2D segmentation maps. 5. Conducting comprehensive evaluations on various public and private datasets to thoroughly assess the effectiveness of our proposed method. 5. Conducting comprehensive evaluations on various public and private datasets to thoroughly assess the effectiveness of our proposed method.

1.2. Contributions

We propose a novel label-efficient disentangled learning approach that addresses the limitations of the current state-of-the-art for retinal

layer segmentation. In particular, we significantly extend and validate SD-LayerNet on a wide range of tasks. Our main contributions are as follows:

- A novel fully differentiable topological engine that converts boundary positions into pixel-wise 2D segmentations, enabling corrections to boundary positions before conversion and ensuring strict topological guarantees on the 2D segmentation maps, and enabling the simultaneous use of both 1D and 2D representations of retinal layers for network training
- A simplified set of retinal anatomical priors, encoded as loss terms, that streamline the formulation and improve robustness to scans with significant tilt, while maintaining the performance of our previous approach in semi-supervised settings and with limited labeled data.
- Extensive evaluation of our method on a total of five private and public datasets covering three retinal diseases, and including the image domain shift across different OCT devices.

2. Methods

Our model (Fig. 2), segments retinal layers and boundaries in OCT images using both labeled and unlabeled data. In particular, it uses a disentangled representation, which is a combination of spatial *anatomical*, and non-spatial *style* factors, to reconstruct the input image using a specialized decoder. A U-Net-based anatomical encoder predicts both the initial surface positions in 1D and a textural factor encoding the noise present in the images. Bespoke *anatomical priors* guide the anatomical encoder to regress anatomically plausible positions. The topological engine converts the regressed surface positions into 2D layer representations to form the binary anatomical factors. These factors, along with style factors that encode the intensities of the spatial factors, make up the *disentangled representation* from which the input image is reconstructed. The overall cost function is a combination of the self-supervised anatomical priors, the reconstruction loss, and the supervised loss of the regression and segmentation task.

2.1. Anatomy encoder

A U-net with five decoder stages and an EfficientNet-b4 encoder is employed for generating the anatomical factors. This anatomical encoder has two output branches: *conv-l* (Fig. 2) for regressing the initial surface positions using a probability map followed by soft-argmax [23], and *conv-t* for generating the texture factor. The texture factor branch produces a single output channel of the same size as the input, that is activated using the Sigmoid function. It is designed to capture the speckle noise present in the scans.

2.1.1. Layer position regression

For each B-scan that is segmented into S surfaces, the *conv-l* branch of the neural network estimates a position distribution map $P_{W \times H \times S}$, where W and H are the width and height of the B-scan, respectively. This map is created using a column-wise (A-scan) softmax function [23]: for each A-scan, a probability mass function $P(Y | i, s)$ that represents the probability of $Y = r$ on the A-scan i for the surface s is inferred, and r denotes a row number.

The position y_i^s at the A-scan i of the surface s is then determined as the expected value of Y , $y_i^s = \sum_r r \cdot P(r | i, s)$. Similarly to [23,38], we enforce the correct topological order of the layers by iteratively updating the predicted boundary positions:

$$y_i^s = y_i^{s-1} + |y_i^s - y_i^{s-1}|_+ \quad (1)$$

where $|\cdot|_+$ denotes the non-negative part of the absolute value.

In the training phase, when manual annotations are available, we minimize two loss terms to guide the network towards a correct layer boundary segmentation. First, the mean smooth L1 loss is minimized

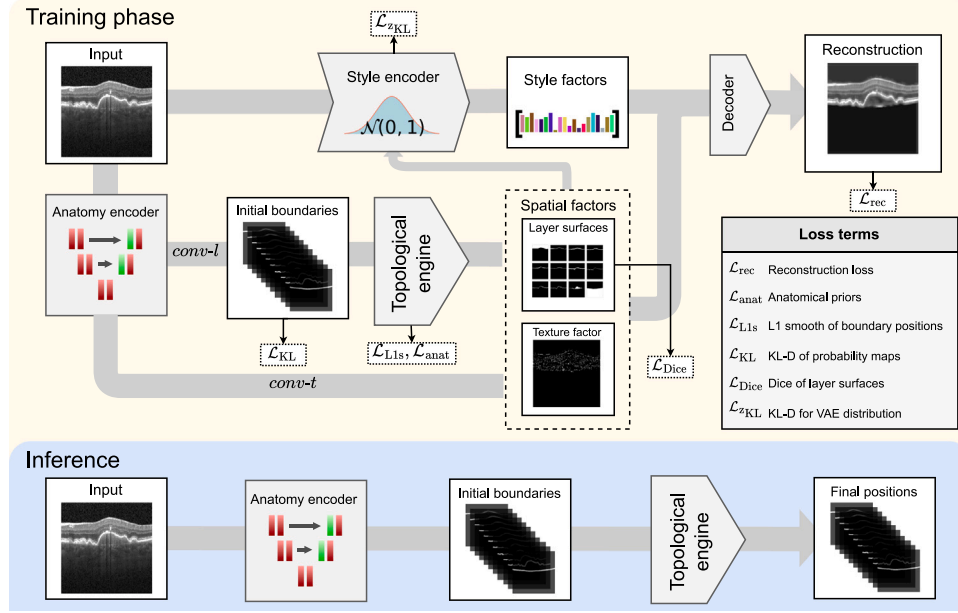


Fig. 2. The model consists of two input branches. Initially, an anatomy encoder yields two output branches for feature extraction. One of these outputs functions directly as a sole spatial texture factor, while the other generates a probability map outlining layer positions for regression. An underlying topological mechanism ensures the accurate sequence of layers and generates spatial maps. The second input branch is linked to a style encoder, producing style factors that encode intensity values of anatomical features. The decoder then reassembles a reconstructed image using these style and anatomical factors. Depending on the presence of layer annotations for the input image, the network minimizes either only a supervised loss or a self-supervised loss as well. At inference time the style encoder, the reconstruction decoder and the spatial factors are not used. The output of the model is the set of the layer boundary positions predicted by the anatomy encoder, with the topological guarantees enforced, as well as the segmentation masks of the layers.

according to the following equation:

$$\mathcal{L}_{L1s} = \frac{1}{W \cdot S} \sum_s \sum_i L_{1s}(y_i^s - \mu_i^s) \quad (2)$$

where μ_i^s represents the reference standard coordinate for the surface s at the A-scan i . The smooth L1 loss is defined as follows:

$$L_{1s}(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } |a| \leq 0.5, \\ |a| - \frac{1}{2}, & \text{otherwise.} \end{cases} \quad (3)$$

Note that the smooth L1 loss is less sensitive to outliers than the mean squared error (MSE) loss, and it may help prevent exploding gradients during training [39]. Moreover, given the limitations of manual annotations, sub-pixel accuracy is often not achievable, making errors smaller than 1 pixel less relevant. This loss function has also been used effectively in previous studies, such as [23,25,30].

Second, we model $P(Y | i, s)$ as a normal distribution. The mean of this distribution is the reference standard position of the surface, denoted as μ_i^s , and the standard deviation is a hyperparameter, denoted as σ . We use the Kullback–Leibler (KL) divergence [40] calculated on the whole B-scan and all boundaries (where being available in the reference annotations) as a loss term during training to encourage the network to learn the target distribution, as suggested in previous works [23,38,41].

$$\mathcal{L}_{KL} = \sum_s \sum_r \sum_i T(r | i, s) \ln \frac{T(r | i, s)}{P(r | i, s)} \quad (4)$$

where $T(Y | i, s)$ represents the target probability map containing a Gaussian normal distribution with parameters μ_i^s and σ .

2.1.2. Topological engine

We introduce a novel method (Fig. 3) for creating a 2D representation from a 1D representation of a layer. Instead of using the top-down cumulative sum approach from [37], we directly generate the 2D representation via a matrix M , defined such that each element $M[i, j]$ is equal to the row number r . We subtract the corrected layer position vector Y from M , which results in a matrix that is close to

$$\begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 256 & \cdots & 256 \\ \vdots & \vdots & \vdots \\ 512 & \cdots & 512 \end{pmatrix} - \begin{pmatrix} 223 \\ 224 \\ \vdots \\ 256 \end{pmatrix}^T = \begin{pmatrix} -223 & \cdots & -256 \\ \vdots & \vdots & \vdots \\ 33 & \cdots & 1 \\ \vdots & \vdots & \vdots \\ 289 & \cdots & 257 \end{pmatrix} \rightarrow$$

$$\rightarrow \sigma(\cdot) \rightarrow \begin{pmatrix} 0.0 & \cdots & 0.0 \\ \vdots & \vdots & \vdots \\ 0.9 & \cdots & 0.7 \\ \vdots & \vdots & \vdots \\ 1.0 & \cdots & 1.0 \end{pmatrix} \Rightarrow$$

Fig. 3. Illustration of the conversion of the 1D representation to 2D representation.

zero around the predicted position, and negative above it or positive below it. We apply the sigmoid function to this matrix to create a binary representation of the upper and lower portions of the surfaces.

To create mutually exclusive spatial maps for the retinal layers, we iteratively decompose the surface maps into layer maps by subtracting one layer map from the previous one. We round the layer maps' values and the texture factor either to 0 or 1, to form the anatomical factors of the disentangled representation, denoted as F_a . It is important to note that due to the rounding being a non-differentiable operation, it is only used in the forward pass; in the backward pass, the original gradients are used. In the training phase, for the supervised samples, we calculate the Dice loss between the anatomical factors F_a and the pixel-level reference annotations R :

$$\mathcal{L}_{Dice} = \frac{1}{|R|} \sum \frac{2 \cdot |F_a \cap R| + \epsilon}{|F_a| + |R| + \epsilon} \quad (5)$$

where term ϵ is a small constant added to the numerator and denominator to prevent division by zero. The pixel-level annotations are generated from the boundary annotations and for each A-scan, they are defined by the upper and lower boundaries of the layers. $\mathcal{L}_{Dice}$ is

applied selectively only to A-scans where both the upper and lower boundaries of the layer are defined in the reference annotation.

2.2. Style encoder

Following the modality factor generation described in Chartsias et al.'s work [36], we use a Variational Autoencoders (VAE) to learn the posterior distribution of latent factors for input B-scans with the corresponding spatial anatomical factors s . VAEs consist of an encoder and a decoder, which are trained to learn a compact latent space and minimize reconstruction error and the KL-divergence between the estimated distribution in each dimension of the latent space and the prior unit Gaussian distribution. We refer to the latter as $\mathcal{L}_{z_{KL}}$ in accordance with Chartsias et al. [36]. During inference, we sample a stochastic vector from the learned posterior distribution to yield the final style factor F_s .

2.3. Image decoder

In order to guide the network to predict realistic layers and layer boundaries, the spatial anatomical factors F_a and the non-spatial style factors F_s are used to reconstruct the original image. As adjacent layers typically have different intensity values, the network is encouraged to correctly delineate the layer boundaries to achieve the best-possible reconstruction.

The reconstruction process is carried out using a decoder convolutional network that is conditioned with four FiLM layers [42]. FiLM is a general-purpose conditioning method that learns scale and offset parameters for each anatomical factor channel within a convolutional architecture using the style factors F_s as input parameters. With this design, the information stored in F_s can only influence the intensity levels of the generated image. This feature is crucial in the effort to separate out the spatial and style factors, as it encourages F_s to not include any details about the spatial structure or layout of the anatomy [36].

The reconstruction loss, represented by $\mathcal{L}_{rec}$, is a measure of the mean absolute error (MAE) between the intensity values of the original and reconstructed images in the region between the predicted ILM and BM surfaces, which are located at the top and bottom of the retina. This spatial restriction is used to eliminate the effect of any noise or uncertainty present in areas outside the retina.

2.4. Anatomical priors

To promote the segmentation of anatomically coherent surfaces, we introduce a set of tailored constraints. These constraints can be utilized even in the absence of annotations, and are especially useful in cases with a scarce amount of training data.

Correct topological order. To reflect the strict ordering of retinal layers, we aim to minimize the mean number and magnitude of topological ordering violations among A-scans. This is achieved through the use of the following topological ordering constraint:

$$\mathcal{L}_{to} = \frac{1}{W} \sum_s \sum_i |*|y_i^{s-1} - y_i^s| \quad (6)$$

The calculation of the loss value occurs prior to the application of the topological ordering operation defined in Eq. (1) on the predicted boundary positions y_i^s .

Elimination of rapid shifts of the layer boundaries. In order to eliminate anatomically implausible rapid fluctuations in the predicted layer boundaries and capitalize on the inherent curvature of distinct retinal layers, we propose constraints on the second-order derivatives of

the layer boundaries. Unlike previous methods, such as the slope and continuity constraint introduced in [37], which are heavily influenced by the inclination of the B-scans, the second-order derivative is much less affected by the orientation of the retina, while it approximates reasonably well the curvature of the layer boundaries.

Since we predict the positions of the layer boundaries only at discrete A-scan locations, where $f_s(x) = y_x^s$, we approximate the second derivative with finite differences:

$$f_s''(x) \approx \frac{-f_s\left(x + \left\lfloor \frac{1}{2}h \right\rfloor\right) + 2f_s(x) - f_s\left(x - \left\lfloor \frac{1}{2}h \right\rfloor\right)}{h}, \quad (7)$$

where hyperparameter h is the window width for the second derivative approximation.

To encourage the shape of the predicted layer boundaries to be within the anatomically plausible limits, we introduce a new loss term $\mathcal{L}_{rs}$, defined as:

$$\mathcal{L}_{rs} = \frac{1}{W \cdot S} \sum_s \sum_i |*|f_s''(i) - R_{s+}| \quad (8)$$

where R_s is the pre-calculated maximum second-order derivative of the layer boundary s , determined from the training set.

As an alternative, we considered the layer curvature measure proposed in Fazekas et al. [41] for limiting the rapid shifts in the layer boundaries, but we found that for this use case, the second-order derivative is more beneficial, due to being less dependent on the window size.

2.5. Overall cost function

The overall loss function $\mathbf{L}$ is a composition of the anatomical priors $\mathcal{L}_{anat} = \lambda_{to}\mathcal{L}_{to} + \lambda_{rs}\mathcal{L}_{rs}$, style encoder loss $\mathcal{L}_{z_{KL}}$, the reconstruction loss $\mathcal{L}_{rec}$, and the supervised loss $\mathcal{L}_{sup} = \lambda_{L1}\mathcal{L}_{L1} + \lambda_{KL}\mathcal{L}_{KL} + \lambda_{Dice}\mathcal{L}_{Dice}$:

$$\mathbf{L} = \mathcal{L}_{anat} + \lambda_{z_{KL}}\mathcal{L}_{z_{KL}} + \lambda_{rec}\mathcal{L}_{rec} + \mathcal{L}_{sup} \quad (9)$$

where λ_* denote weighting terms. The $\mathcal{L}_{sup}$ term is set to 0 for items without annotations.

2.6. Inference time

The style encoder, the spatial factors and the reconstruction components, along with the anatomical priors encoded as loss terms, are primarily designed to facilitate the semi-supervised training. At inference time, these components are not used. Instead, the final result of the model consists of the set of layer boundary positions, and the segmentation masks of the layers, both with the topological constraint described in Eq. (1) enforced.

3. Results

3.1. Experimental setup

3.1.1. Datasets

Our experiments were conducted using three publicly available and two internal datasets.

Public datasets. HCMS dataset from Johns Hopkins University [38] consists of 14 healthy controls (HCs) and 21 people with multiple sclerosis (MS). Each subject has 49 B-scans, which were acquired using a Heidelberg Spectralis scanner and nine surfaces manually labeled. The scans are 496×1024 in size and have a lateral resolution of $5.8 \mu\text{m}$, an axial resolution of $3.9 \mu\text{m}$, and a B-scan distance of $123.6 \mu\text{m}$. The dataset was split into a training set of 735 scans from 6 healthy controls and 9 individuals with MS, and a test set of 980 scans from the remaining 20 individuals. The training set was further divided into a validation set consisting of scans from 2 healthy controls and 2 individuals with MS. The data split and validation set were based on those used in the previous study by the authors of the dataset [23]. In

Table 1

Overview of the model sizes and computational costs of the baseline models and our proposal.

Model	Parameter count (millions)	GFLOPs
ReLayNet [19]	19.5	190
He et al. [23]	19.7	900
TRLS [25]	0.18	10
UNeXt [21]	1.5	8
MGU-Net [20]	4.4	34
TCCT [22]	21.3	233
Ours	20.8	979

our experiments, we used the unflattened version of the dataset.

DukeDME dataset [7], consists of 110 B-scans with a resolution of 496×768 from 10 diabetic macular edema (DME) patients, each with a total of 61 B-scans. In each volume, 11 B-scans contain annotations for 8 retinal layers and for macular edema from a senior and a junior ophthalmologist. However, unlike the HCMS dataset, the external limiting membrane (ELM) is not labeled in this dataset. To adapt the network for this dataset, we modified the final output channel of the *conv-l* branch to 8, but kept the other hyperparameters unchanged. We used a 50%–50% train/test split (as in Chiu et al. [7]; Rathke et al. [13]; Karri et al. [43] and Roy et al. [19]) trained the network on the last 5 patients (55 B-scans) with mild edema until convergence, subsequently, we then tested the model on the first 5 patients (55 B-scans), which have severe edema.

RETOUCH dataset [44] has been used as an unsupervised training set in a few experiments. This dataset includes 112 OCT volumes from 112 patients, which were acquired using SD-OCT devices from three different manufacturers: Cirrus HD-OCT (Zeiss Meditec), Spectralis (Heidelberg Engineering), and T-1000/T-2000 (Topcon). Half of the patients in the dataset were diagnosed with macular edema due to age-related macular degeneration (AMD), while the other half had edema due to retinal vein occlusion (RVO). The dataset includes pixel-level annotations for intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelial detachment (PED), but does not have annotations for layer boundaries. Therefore, we did not utilize any of the available annotations in our work. For our experiments, we used 24 Spectralis volumes from the training set. Each volume consists of 49 B-scans with a resolution of 512×496 pixels, covering a macular area of $6 \times 6 \text{ mm}^2$ with an axial resolution of $4.9 \mu\text{m}$.

Private datasets. Given the lack of large, publicly available datasets featuring severely damaged retinal structures, we also utilized two internal datasets from the Department of Ophthalmology, Medical University of Vienna (MUV) containing OCT scans of patients undergoing treatment for AMD. The first dataset was acquired using Spectralis (Heidelberg Engineering) and comprises voxels with dimensions ranging from $(512 \times 19 \times 496)$ to $(1024 \times 49 \times 496)$, covering a volume of $6 \times 6 \times 2 \text{ mm}^3$. This dataset includes intermediate and late-stage wet-AMD cases from real-world clinical settings and consists of 68 volumes. Within these volumes, 7 uniformly spaced B-scans contained manual layer annotations by experts. This amounts to a total of 476 annotated B-scans. This is the same dataset as was used in Fazekas et al. [37].

The second dataset from MUV was acquired using Cirrus (Zeiss Meditec) and consists of 66 volumetric scans with voxel dimensions of $512 \times 96 \times 512$, spanning a volume of $6 \times 6 \times 2 \text{ mm}^3$. We divided this dataset into a training set (40%), a validation set (10%), and a test set (50%). As an *unsupervised Cirrus dataset*, we used a third dataset from the same patient cohort, consisting of 33 volumes from 33 patients without any layer annotations.

3.1.2. Experimental settings

Data preparation. For training, we use random patches of size 512×512 . In cases where the vertical resolution of the B-scan was smaller than 512, zero-padding was applied to extend it to the desired dimensions. Each individual B-scan was normalized to achieve zero mean and

unit standard deviation. Throughout the training phase, B-scans were subject to random horizontal flipping with a probability of 50%. During validation and testing, complete B-scans were supplied to the networks after undergoing B-scan-wise normalization.

Evaluation metrics. We adopted Mean Absolute Difference (MAD) as our metric of choice because it offers an unbiased assessment, treating all segmentation errors equally. This suits our layer boundary regression task, ensuring a balanced evaluation that considers both minor and major discrepancies between automated and manual delineations.

Evaluation of pixel-level classification baseline methods. Since the MAD is calculated based on layer boundaries, it was necessary to post-process the output of layer segmentation methods that produce pixel-level classification in order to derive these boundaries. To ensure fairness in our evaluation, we applied the following algorithm to the pixel-level segmentations:

First, we removed all connected components that were entirely above the Inner Limiting Membrane (ILM) in the reference annotation, filtering out clearly incorrect segmentations. Next, for each layer class in each A-scan, we identified the boundary position as the vertical location of the first occurrence of that layer. If a layer class was absent in an A-scan, we assigned its boundary position based on the position of the next detected layer, assuming that the absent layer had collapsed from top to bottom.

Implementation and training details. Our model was trained using mixed-precision on an Nvidia GeForce RTX 2080 Ti GPU with an Intel Xeon Silver 4114 CPU, with 16 cores, using Python 3.8.8 and PyTorch 1.8.1 on a CentOS 7 operating system. To determine the weighting terms for the loss function, we performed a grid search on the training and validation set, testing values in the range of 0.01 to 10. Our experiments revealed that the hyperparameter settings were largely insensitive to variations within this range. Consequently, all loss weighting terms, λ_* , were set to 1 in our experiments, except for λ_{rec} , which was set to 0.1. Notably, assigning a higher weight to the reconstruction loss negatively impacted segmentation performance. The source code for our implementation will be made available at <http://github.com/ABotond/SD-LayerNet2>.

3.1.3. Experiments

We conducted a comprehensive *baseline study* in which we compared the performance of our method against multiple state-of-the-art segmentation models. Subsequently, we conducted a series of ablation studies (*Ablation studies*) to thoroughly analyze the effectiveness of the various components within our method. Additionally, we evaluated the method's performance in various real-world scenarios. Specifically, we assessed its performance when only a limited number of labeled samples are available for training. Furthermore, we investigated the impact of the choice of unlabeled datasets on the model's performance. We also studied how the method behaves under image domain shift by evaluating it on datasets acquired with an OCT device it was not trained on. An overview of the experiments and the corresponding datasets can be found in Table 2.

Baseline studies. We compared our proposed method with several baselines: ReLayNet, MGU-Net, UNeXt, TCCT-BP, TRLS and the model proposed by He et al. [23]. ReLayNet [19] is a dedicated pixel-wise retinal OCT layer segmentation network, featuring a final layer comprising channels for each layer and the background. UNeXt [21] employs a similar output architecture but achieves comparable performance with significantly fewer parameters. MGU-Net [20] consists of two sequential networks. The first network generates an initial segmentation of the retina, aiding the second network in predicting a more refined pixel-wise segmentation of the retinal layers. TRLS [25] directly regresses the retinal layer boundary positions with a transformer based architecture. As we were unable to reproduce the results with this method, we did not conduct statistical tests with this method and

Table 2
Overview of the experiments and the datasets used.

Experiment	Dataset				
	HCMS [23]	DukeDME [7]	RETOUCH [44]	Internal MUV Spectralis	Internal MUV Cirrus
Baseline studies	•	•	•	•	•
Ablation studies	•				
Label efficiency	•				
Impact of unsupervised dataset	•	•	•		
Domain shift				•	•

Table 3
Overview of the ablation studies.

Method name	Loss terms			
	$\mathcal{L}_{rec}$	$\mathcal{L}_{anat}$	$\mathcal{L}_{L1s} + \mathcal{L}_{KL}$	$\mathcal{L}_{Dice}$
He et al. [23] + $\mathcal{L}_{anat}$		•	•	•
Ours - $\mathcal{L}_{anat}$	•		•	•
Ours - ($\mathcal{L}_{L1s} + \mathcal{L}_{KL}$)	•	•		•
Ours - $\mathcal{L}_{Dice}$	•	•	•	
SD-LayerNet	•	•	•	•

reported the results presented in Cao et al. [25], but converted to μm . TCCT [22], the winner of the 2022 MICCAI GOALS layer segmentation challenge [45] utilizes a hybrid transformer-CNN architecture for effectively segmenting layers. The state-of-the-art model proposed by He et al. [23] incorporates two output branches. One branch provides pixel-wise layer segmentation, while the second branch yields a regression of the layer boundary positions with guarantees on the correct topological order. For consistency, we utilized the original source code for ReLayNet, UNeXt, and MGU-Net, training these models while using the same data augmentations as for SD-LayerNet. For He et al.'s method, we reimplemented it in-house and performed thorough validation on both the HCMS and DukeDME datasets. An overview of the parameter sizes and computational costs can be found in Table 1.

Ablation studies (Table 3). A series of ablation studies were undertaken using the HCMS dataset. These experiments were performed on both the full training set and reduced subsets, with the left-out training data serving as the unsupervised training set in each case. We opted for the HCMS dataset as it has high-quality scans with minimal pathological anatomical deformations, allowing to effectively study the impact of our contributions for cases where existing methods are already performing adequately.

In the first ablation study (He et al. + $\mathcal{L}_{anat}$), we assess the effect of the proposed disentangled approach, in contrast to solely using the unsupervised anatomical priors. We converted the layer segmentation method developed by He et al. [23] into a semi-supervised method using additional unsupervised data and incorporating our anatomical priors. In the second ablation study, we trained SD-LayerNet without the use of anatomical priors (Ours - $\mathcal{L}_{anat}$), relying only on the reconstruction as a source of loss in the unsupervised setting. In the third ablation study, we evaluated the importance of the coupled 2D surface representation by conducting experiments with only the Dice loss for supervision (Ours - ($\mathcal{L}_{L1s} + \mathcal{L}_{KL}$)) and with only smooth L1 loss and KL-divergence as the loss function (Ours - $\mathcal{L}_{Dice}$).

Label efficiency. To assess the performance of the methods with a limited number of labeled samples, we gradually reduced the number of patients included in the HCMS training set from 15 down to 1 while keeping all the annotated B-scans for a given patient. The removed patients were used without annotations as the unsupervised training set. The size of the test set remained unchanged. We then used a linear mixed-effect model test [46] to evaluate the results, taking into account the interdependency of the B-scans within the same volume.

Semi-supervised performance. To investigate and better understand the impact of the choice of the unsupervised data on performance, we conducted several experiments using the DukeDME dataset and various unlabeled samples from different datasets. In particular, we aimed to verify the impact on the performance of the model when adding unlabeled: i. highly correlated, but unannotated B-scans from the same dataset (DukeDME), ii. good-quality scans from the HCMS dataset (without the annotations) featuring HC or MS patients, and iii. the RETOUCH dataset with macular edema secondary to RVO or AMD, rather than DME.

Robustness to image domain shift. We conducted experiments to assess the robustness of our model to image domain shift. We conducted four experiments using our internal MUV datasets with Cirrus and Spectralis OCTs, and compared the results with the best-performing baseline method:

- **100% Cirrus:** we trained a model on the fully labeled Cirrus training set, along with the additional unsupervised Cirrus dataset. This emulates a scenario where there is a considerable amount of training data available for one particular type of OCT device and serves as one of the baselines for this domain shift study.
- **100% Cirrus + Spectralis:** we trained the model on the combined full Cirrus and Spectralis training set. This simulated a situation where abundant training data is available for different domains simultaneously. This experiment served as the second baseline for our domain shift study.
- **10% Cirrus:** we reduced the number of annotated Cirrus scans to 10% (corresponding to 5 patients), while retaining all the Spectralis and unsupervised Cirrus scans in the training set. This setup aimed to replicate a scenario where there is sufficient training data for one type of OCT device but limited labeled data for a specific device.
- **0% Cirrus:** we completely excluded annotated Cirrus scans from the training set. Instead, our method was trained solely using annotated Spectralis scans and unsupervised Cirrus scans.

3.2. Results

3.2.1. Qualitative and baseline study results

The qualitative results obtained from the DukeDME dataset (Fig. 4) reveal certain limitations of the pixel-level classification methods, as they often introduce topological errors in the segmentations. These errors manifest as discontinuities or even the complete absence of part of certain layers by classifying them as background or as a different layer. On the other hand, the boundary regression-based method developed by He et al. exhibits more consistent results, although it may deviate significantly from the ground truth in certain areas (Fig. 4, first row). In contrast, our models demonstrate a more robust segmentation overall, but they do encounter difficulties in areas with high fluid concentration and missing layers. The quantitative results on the DukeDME dataset (Table 4) underline these findings, by showing a 12% improvement over the best-performing baseline method while being statistically significantly better than all the state-of-the-art methods tested.

Table 4

Quantitative results on the DukeDME dataset. The values correspond to the mean absolute distance (MAD) metric.

Method	ILM	RNFL-GCL	IPL-INL	INL-OPL	OPL-ONL	ISM-ISE	OS-RPE	BM	Total
ReLayNet [19]	15.72 ± 19.46 ^{***}	13.96 ± 8.70 ^{***}	12.47 ± 9.45	13.36 ± 8.59	12.49 ± 3.98 ^{***}	8.38 ± 3.34 ^{***}	7.55 ± 1.56 ^{***}	12.24 ± 8.99 ^{***}	12.01 ± 6.29 ^{***}
TRLS [25]	14.68 ± 33.5	18.14 ± 31.68	15.42 ± 28.22	16.09 ± 28.76	18.01 ± 29.16	6.28 ± 8.47	4.64 ± 6.79	5.38 ± 8.84	12.33 ± 21.94
UNeXt [21]	7.89 ± 4.06 ^{***}	8.65 ± 1.52 ^{***}	9.37 ± 2.14 ^{***}	9.50 ± 2.06	10.44 ± 3.20	5.22 ± 1.15 [*]	4.30 ± 0.45	4.35 ± 0.86	7.37 ± 0.93 ^{***}
MGU-Net [20]	4.78 ± 1.33	8.85 ± 2.75 ^{***}	8.64 ± 1.90 ^{***}	11.09 ± 3.48 ^{***}	11.18 ± 3.36	4.83 ± 1.09 [*]	4.83 ± 1.06	5.74 ± 1.22 ^{***}	7.34 ± 1.08 ^{***}
TCCT [22]	4.63 ± 1.11	8.03 ± 1.79 ^{***}	7.74 ± 1.90 ^{***}	9.63 ± 6.65 ^{***}	9.38 ± 3.01	4.83 ± 1.09 [*]	4.95 ± 1.32	5.12 ± 1.02 ^{***}	7.04 ± 1.10 ^{***}
He et al. [23]	4.86 ± 0.79 ^{***}	8.06 ± 2.76 ^{***}	7.72 ± 1.94	8.77 ± 2.40	9.55 ± 2.62	5.01 ± 1.76 [*]	5.13 ± 1.19 ^{***}	5.36 ± 1.30 ^{***}	6.71 ± 0.92 ^{***}
Ours with Corr. scans	4.62 ± 0.86	7.65 ± 1.92 ^{***}	7.55 ± 1.65	8.61 ± 2.30	8.93 ± 2.77	4.08 ± 0.82	4.53 ± 0.41	4.35 ± 0.63	6.19 ± 0.86
Ours with HCMS	4.56 ± 0.66	6.55 ± 1.23	7.24 ± 2.24	8.34 ± 2.70	8.57 ± 2.46	4.39 ± 1.23	4.30 ± 0.74	4.07 ± 0.41	5.91 ± 0.90
Ours with RETOUCH	4.40 ± 0.68	6.49 ± 1.21	7.10 ± 1.92	<u>8.53 ± 2.75</u>	<u>8.76 ± 3.36</u>	4.71 ± 1.82	<u>4.53 ± 0.85</u>	4.98 ± 1.13 ^{***}	<u>6.09 ± 1.20</u>

No statistical tests were conducted on the TRLS model.

Corr. scans stays for correlated scans.

^{*} Indicates statistically significant differences to Ours + Corr. scans.^{**} Indicates statistically significant differences to Ours + HCMS.^{***} Indicates statistically significant differences to Ours + RETOUCH.

Table 5

Quantitative results on the HCMS dataset, including the ablation experiments. The values correspond to the mean absolute distance (MAD) metric.

Method	ILM	RNFL-GCL	IPL-INL	INL-OPL	OPL-ONL	ELM	IS-OS	OS-RPE	BM	Total
ReLayNet [19]	4.75 ± 3.02 [*]	5.27 ± 4.98 [*]	5.85 ± 5.36 [*]	6.03 ± 5.34 [*]	5.37 ± 5.81 [*]	5.52 ± 6.43 [*]	4.42 ± 5.44 [*]	4.97 ± 5.09 [*]	4.07 ± 3.60 [*]	5.14 ± 4.83 [*]
TRLS [25]	3.19 ± 1.76 [*]	4.72 ± 2.29 [*]	3.79 ± 1.33 [*]	4.11 ± 1.09 [*]	3.38 ± 1.21 [*]	3.59 ± 1.77 [*]	3.27 ± 1.31 [*]	3.86 ± 1.43 [*]	3.55 ± 1.66 [*]	3.72 ± 1.21 [*]
UNeXt [21]	7.26 ± 3.11 [*]	5.66 ± 2.92 [*]	5.10 ± 2.95 [*]	5.35 ± 3.13 [*]	4.16 ± 1.20 [*]	3.77 ± 1.01 [*]	2.99 ± 0.96 [*]	4.01 ± 1.60 [*]	3.70 ± 1.37 [*]	4.67 ± 1.60 [*]
MGU-Net [20]	3.19 ± 1.76 [*]	4.72 ± 2.29 [*]	3.79 ± 1.33 [*]	4.11 ± 1.09 [*]	3.38 ± 1.21 [*]	3.59 ± 1.77 [*]	3.27 ± 1.31 [*]	3.86 ± 1.43 [*]	3.55 ± 1.66 [*]	3.72 ± 1.21 [*]
TCCT [22]	2.73 ± 0.93 [*]	3.87 ± 1.16 [*]	3.22 ± 0.88 [*]	3.91 ± 0.90 [*]	3.45 ± 1.19 [*]	3.12 ± 1.09 [*]	2.44 ± 0.99 [*]	3.75 ± 1.12 [*]	3.05 ± 1.90 [*]	3.22 ± 0.95 [*]
He et al. [23]	2.40 ± 0.31 [*]	3.02 ± 0.72	2.94 ± 0.46 [*]	3.25 ± 0.62 [*]	2.84 ± 0.58 [*]	2.63 ± 0.76	2.09 ± 0.54 [*]	3.52 ± 0.77	3.07 ± 1.67 [*]	2.86 ± 0.37 [*]
He et al. + $\mathcal{L}_{anat}$	2.49 ± 0.57 [*]	2.91 ± 1.00	2.81 ± 0.57	3.10 ± 0.56 [*]	2.87 ± 0.61 [*]	2.63 ± 0.81	1.91 ± 0.69 [*]	3.43 ± 1.10	2.78 ± 2.11	2.77 ± 0.57 [*]
Ours - $\mathcal{L}_{anat}$	2.33 ± 0.31	3.17 ± 1.34 [*]	3.07 ± 0.79 [*]	3.12 ± 0.65 [*]	2.93 ± 0.75 [*]	2.57 ± 0.70	1.86 ± 0.67	3.71 ± 1.74 [*]	2.84 ± 2.11	2.84 ± 0.69 [*]
Ours - $\mathcal{L}_{L1s}$	2.30 ± 0.24	2.85 ± 0.93	2.86 ± 0.59 [*]	3.06 ± 0.55	2.75 ± 0.65	2.54 ± 0.51	1.82 ± 0.65	3.43 ± 1.38	2.90 ± 2.27 [*]	2.72 ± 0.54
Ours - $\mathcal{L}_{Dice}$	2.30 ± 0.24	<u>2.86 ± 0.86</u>	2.94 ± 0.55 [*]	3.17 ± 0.60 [*]	2.88 ± 1.08	2.71 ± 0.86	2.01 ± 0.79	3.33 ± 0.96	2.84 ± 2.03	2.78 ± 0.54
Ours	2.28 ± 0.31	2.95 ± 1.23	2.77 ± 0.38	3.02 ± 0.44	2.72 ± 0.63	<u>2.57 ± 0.77</u>	<u>1.84 ± 0.85</u>	<u>3.40 ± 1.00</u>	2.75 ± 2.15	2.70 ± 0.49

No statistical tests were conducted on the TRLS model.

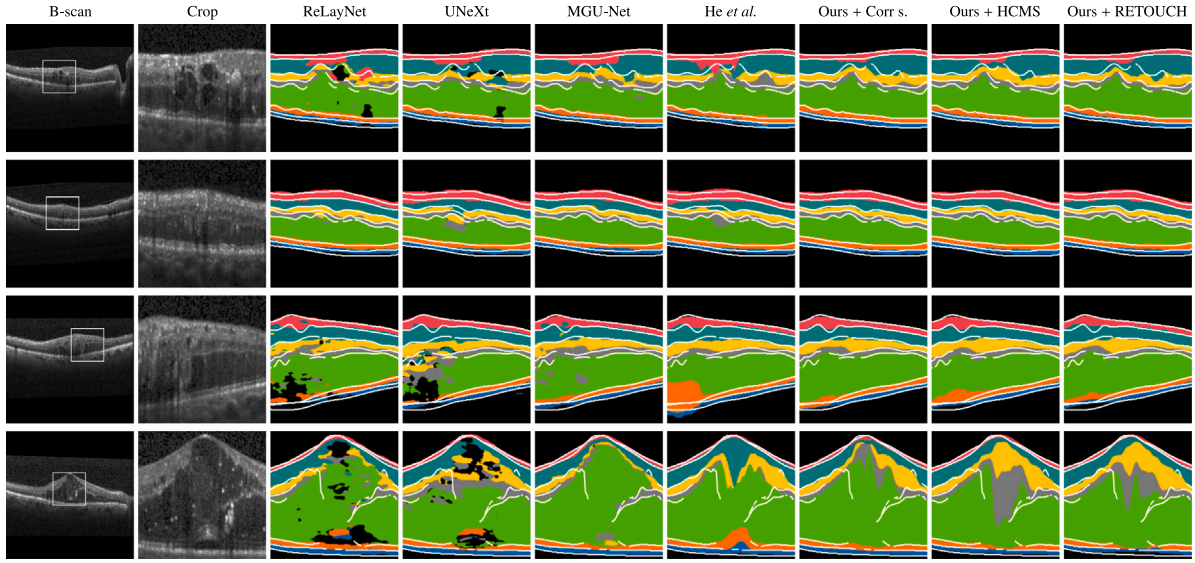
^{*} Indicates statistically significant differences to our method.

Fig. 4. Qualitative results of the baseline methods and our model variants on the DukeDME dataset. The white lines depict the manual annotations, while the colored areas represent the pixel-level classifications.

Additionally, the qualitative results obtained from the HCMS dataset, trained with a single patient (Fig. 5), also exhibit similar artifacts observed in pixel-level classification methods, particularly in challenging areas with vessels. Altogether, our models outperform the layer boundary regression method proposed in He et al. by 5.5%, maintaining robust segmentation performance while ensuring topological consistency (Table 5).

3.2.2. Ablation studies

The ablation studies (Table 5) showed that the inclusion of our proposed anatomical prior loss terms in He et al. resulted in an overall improvement of 3% in the segmentation results. Secondly, we observed

that omitting anatomical constraints altogether (Ours - $\mathcal{L}_{anat}$), and relying solely on the unsupervised reconstruction loss for the self-supervised training resulted in inferior performance, thus showing the importance of these factors for the semi-supervised pipeline. Thirdly, the concurrent use of smooth L1 loss and Dice loss functions led to better performance in most of the layers compared to using either of them separately, showing that the regression and segmentation losses are complementary.

3.2.3. Label efficiency evaluation

Evaluation of label efficiency (Fig. 6) demonstrated that although our proposed method and the baseline method by He et al. [23]

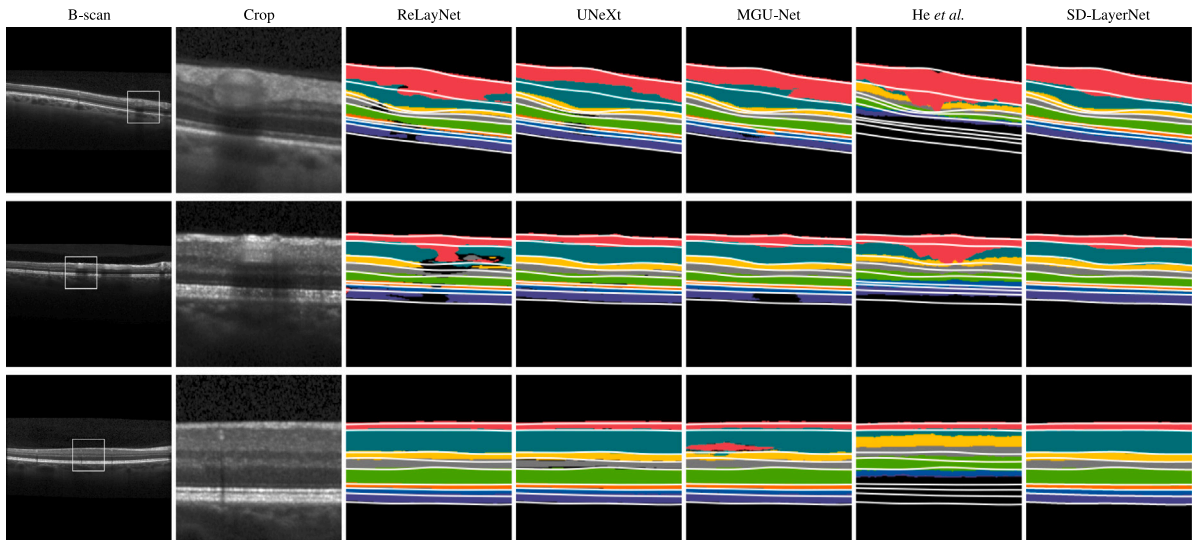


Fig. 5. Qualitative results of the baseline methods and our model on the HCMs dataset, when the models were trained with a single patient only. The white lines depict the manual annotations, while the colored areas represent the pixel-level classifications.

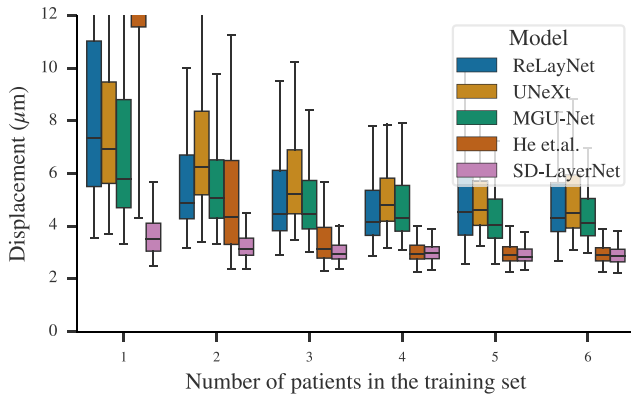


Fig. 6. Quantitative results (MAD) of the label efficiency experiments on all layer boundaries combined in the HCMs dataset.

exhibited similar performance on the relatively simple HCMs dataset, our model is significantly more robust to the amount of labeled data available, excelling in a low labeled data regime. Namely, when the number of labeled patients was reduced to two or one, the performance of our model experienced only a minor deterioration of 5%, while the other methods have shown 2 to 5× worse performance. The qualitative examples (Fig. 5) support this finding, as the proposed model is still able to correctly identify the layers, along with topological consistency.

3.2.4. Semi-supervised performance

Table 4 shows that the choice of the unsupervised dataset has an impact on the model's performance, particularly on the RNFL/GCL and Bruch's membrane. Notably, the addition of highly correlated but unannotated scans showed the lowest impact on improving the segmentation performance. We hypothesize that this is due to the limited amount of new information available in the scans. In fact, in the cases of RNFL/GCL, it even resulted in a statistically significant difference in performance compared to models trained with other unsupervised datasets. We theorize that the reason for this is that this layer is severely damaged in many of the scans in the training set, while mostly intact in the HCMs and RETOUCH datasets. Furthermore, the inclusion of the

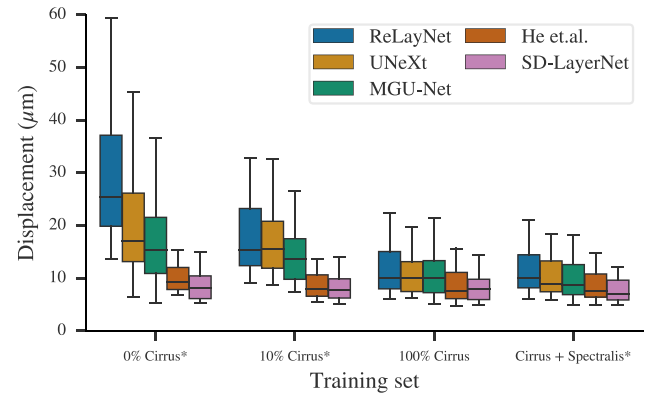


Fig. 7. Image domain shift experiments (MAD) on all layers combined. A model was trained with only Spectralis images (0% Cirrus), with 10% of the Cirrus training set and Spectralis scans (10% Cirrus), with the whole Cirrus training set without Spectralis scans (100% Cirrus), and with the full Cirrus and Spectralis training set (Cirrus + Spectralis). The asterisk indicates statistical significance between He et al. and SD-LayerNet ($p < 0.05$).

RETOUCH dataset, which comprised patients diagnosed with different diseases than those in the DukeDME dataset, negatively affected the segmentation of Bruch's Membrane. This outcome can be attributed to the presence of AMD cases in the RETOUCH dataset, which results in the BM being separated from the RPE, which is not the case in the DukeDME dataset. On the other hand, the inclusion of the HCMs dataset yielded the highest improvement in segmentation performance, possibly due to the good quality of the scans in that dataset. Overall, our results demonstrate a statistically significant enhancement in segmentation performance for our method compared to the baselines. These findings underscore the importance of carefully selecting and considering the composition of the unsupervised dataset when training the segmentation model.

3.2.5. Robustness to image domain shift

The experiments (Fig. 7) demonstrate that our model exhibits superior adaptation capabilities to scans taken by OCT scanners from

different vendors, even when trained with limited or no supervised training samples from those devices. This results in almost no deterioration in performance for our proposed methods, while other methods exhibit almost 2.5× higher error rate. This adaptability can be attributed to leveraging reconstruction and anatomical constraint losses on the unsupervised training data. Remarkably, there were no statistically significant differences observed among the performances of our models trained on the 0% *Cirrus*, 10% *Cirrus*, and 100% *Cirrus* datasets. However, the model trained on the full 100% *Cirrus* + *Spectralis* dataset outperformed the other methods by a statistically significant margin, due to having been trained on a much larger training set. It is worth noting that our method consistently outperformed the baseline method in all dataset versions, except for the 100% *Cirrus* case, where the difference in performance was not statistically significant ($p = 0.08$). For all other dataset versions, our method achieved statistically significant superiority over the baseline methods ($p < 0.05$).

4. Discussion

Our novel layer segmentation model, designed for robustness and label efficiency, surpasses the current state-of-the-art performance across diverse tasks on both public and private datasets. Notably, our innovative fully-differentiable topological engine enables the joint utilization of 1D boundary positions and 2D layers during model training. This approach guides the network in generating distinct, anatomically constrained spatial representations of the layers, facilitating the accurate reconstruction of input images while demanding minimal amount of labels. This not only leads to superior domain adaptation during training but also ensures robust performance in diverse domains, such as scans taken by scanners from different OCT vendors.

Harnessing unlabeled data will be essential in the future of medical AI, particularly given the data-intensive nature of deep learning models. Unlike many non-clinical domains, the annotation of medical data necessitates specialized expertise due to the intricate nature of the tasks and the critical implications of errors. Consequently, only a small proportion of the vast amounts of data generated in clinical settings is annotated. Addressing this challenge, our work advances the field by achieving state-of-the-art performance in retinal layer segmentation using significantly fewer labeled samples, thus demonstrating the potential of leveraging large-scale unlabeled datasets to reduce reliance on manual annotations.

The applications of retinal layer segmentation extend beyond the identification of retinal layer boundaries. Besides the thickness and shape of retinal layers serving as crucial biomarkers, layer segmentation plays an integral role in the OCT angiography (OCTA) analysis, as accurate retinal slab generation relies on precise layer delineation. Importantly, the versatility of our proposed methodology extends beyond retinal layer segmentation, finding relevance in any scenario involving nested anatomical structures and the quantification of tissue thickness (e.g., inner and outer vessel lumen walls, cardiac walls, knee cartilage, etc.).

Despite its strengths, our method is not without limitations. Our experiments show the importance of the selection and composition of the unsupervised training set used during the training of the model, as it can result in statistically significant differences in the segmentation performance of some layers. For practical applications, a broad range of medical conditions and acquisition settings should be covered by the unlabeled training set. Additionally, the regression-based approach always predicts positions for retinal boundaries, even in cases where severe atrophy has effectively eliminated the presence of these boundaries. This introduces a risk of false positives, where boundaries are predicted where none exist, potentially leading to inaccurate assessments and undermining the trustworthiness of the system. While this issue can be partially mitigated through the use of concurrently generated 2D segmentations or by flagging layers with predicted thicknesses below a predefined threshold, an optimal solution would involve the

model explicitly identifying and reporting such instances of boundary absence.

5. Conclusion

Our work presents a novel and robust approach to retinal layer segmentation, which significantly advances the state of the art by leveraging anatomical priors and a fully-differentiable topological engine. This model not only reduces the dependency on labeled data but also ensures strong performance across diverse datasets, including those from different OCT vendors.

In the future, we also plan to expand our research on retinal OCT segmentation to include segmenting lesions while using anatomical priors to ensure a consistent segmentation of the entire retina, further advancing the field's clinical utility. A further direction is to improve the robustness of the methods by automatically detecting layers that cannot be reliably segmented due to extensive damage or imaging artifacts.

Robust retinal layer segmentation methods such as the one presented bear immense clinical utility. They allow clinicians to precisely assess retinal biomarkers, including layer thickness, aiding in early disease detection, monitoring disease progression, and treatment efficacy evaluation. Quantifying these biomarkers through accurate segmentation enhances our understanding of retinal diseases, allowing for tailored patient care and opening new horizons for research and applications in ophthalmology and beyond.

CRediT authorship contribution statement

Botond Fazekas: Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Guilherme Aresta:** Writing – review & editing, Supervision. **Dmitrii Lachinov:** Formal analysis. **Sophie Riedl:** Data curation. **Julia Mai:** Data curation. **Ursula Schmidt-Erfurth:** Resources, Data curation. **Hrvoje Bogunović:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as a potential competing interests:

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